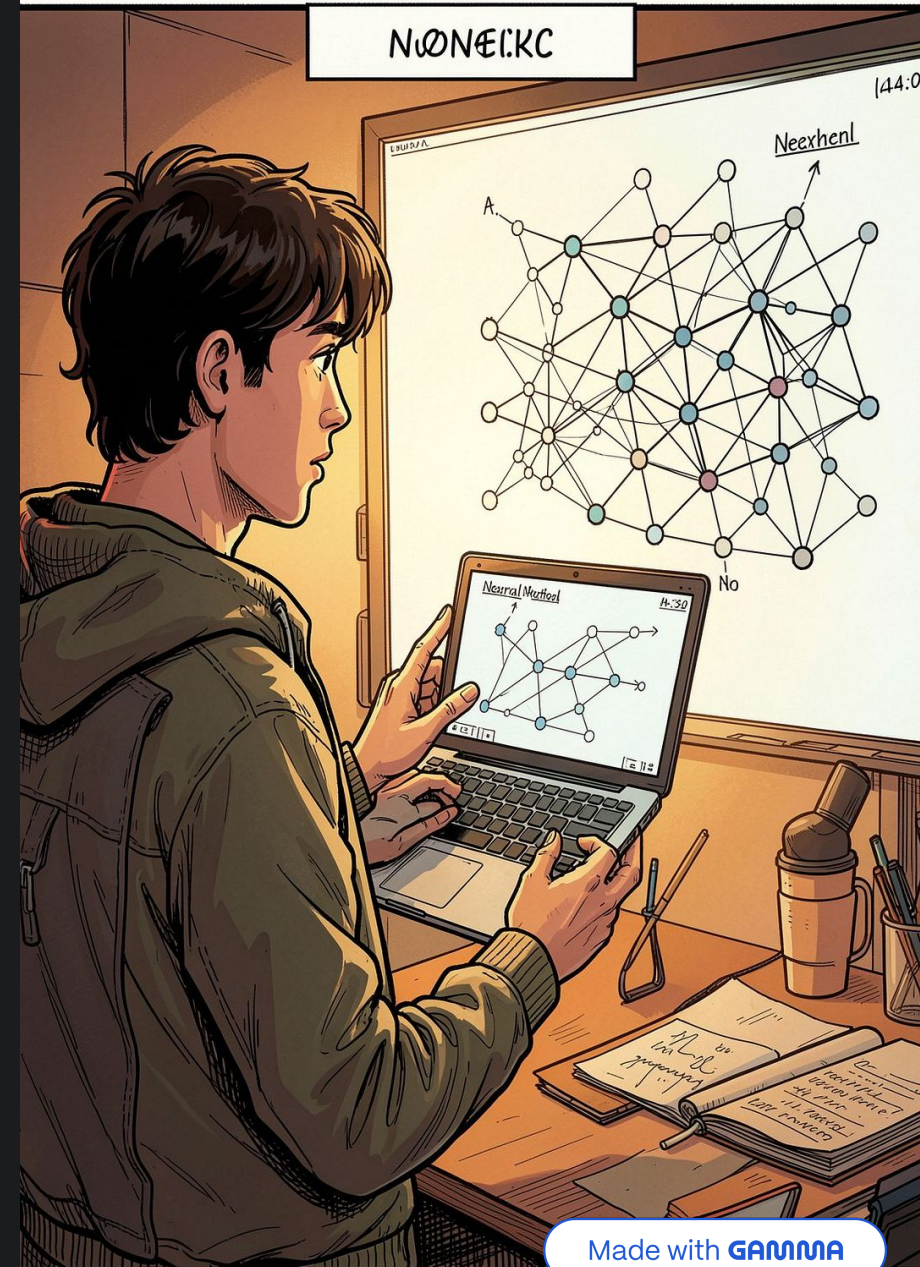


Presented by: Shahid Ali · Roll No: 4148

Introduction to Machine Learning

Machine Learning (ML) is a core area of Artificial Intelligence that enables computers to learn from data to make predictions and decisions used across daily life.



What is Machine Learning?

Definition

Technology that lets computers learn from data and identify patterns.

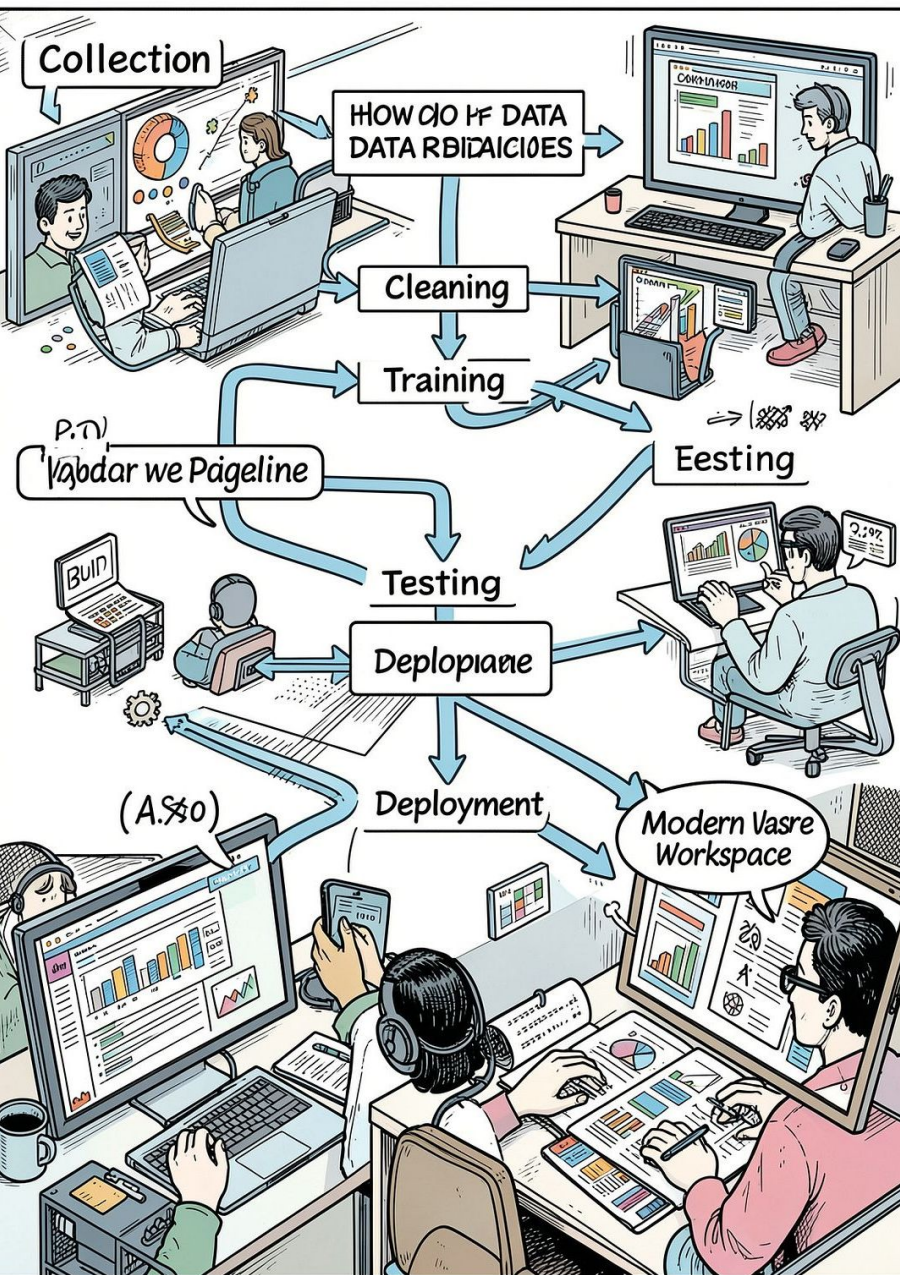
Behavior

Uses patterns to make predictions or decisions without explicit instructions for every task.

Example

Spam detection: learn from examples of spam and normal emails to classify new messages.





How ML Works — Five Steps



Collect Data

Prepare Data

Train Model

Test Model

Make Predictions

Simple example: a house-price model learns from size, location, and past prices to predict new listings.

Types of Machine Learning



Supervised Learning

Learns from labeled data. Example: email classification (spam vs normal).



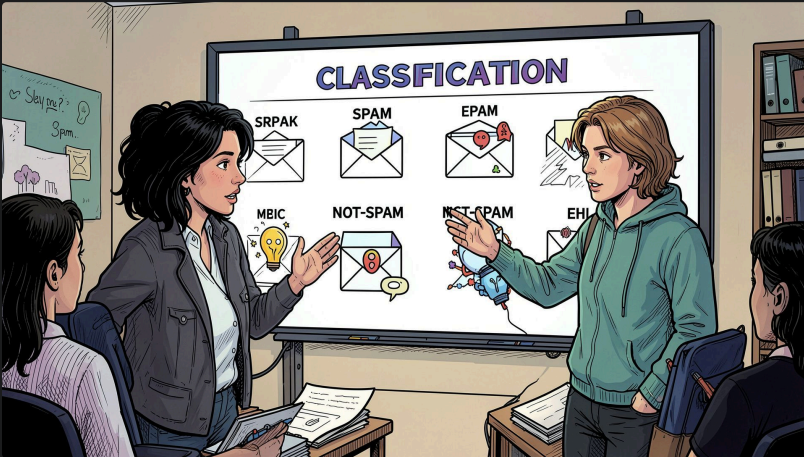
Unsupervised Learning

Finds patterns/groups without labels. Example: customer segmentation.



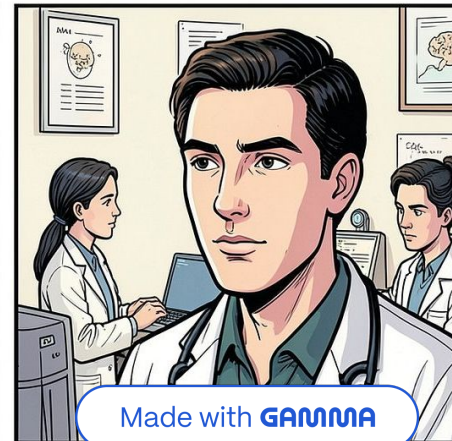
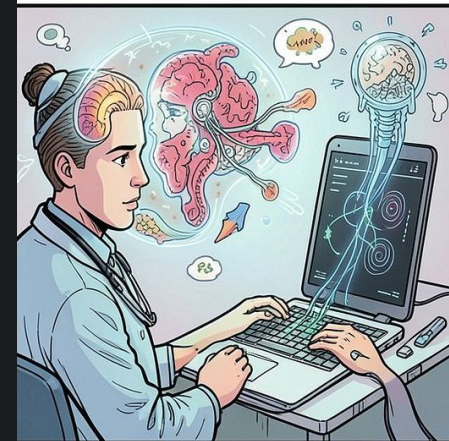
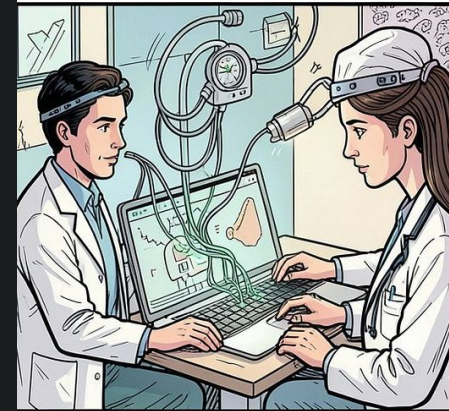
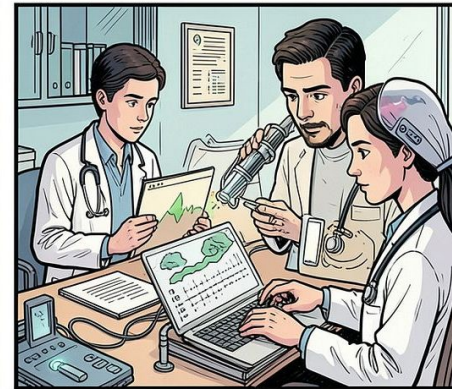
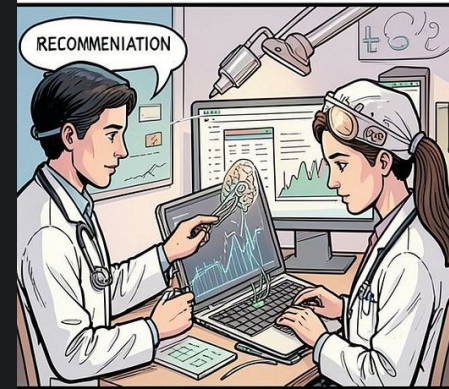
Reinforcement Learning

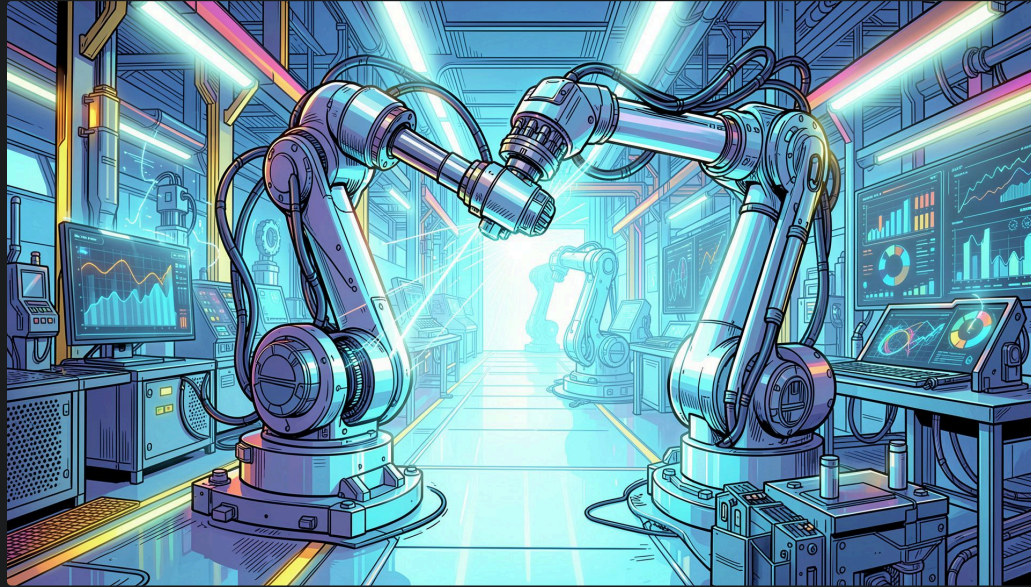
Learns via rewards and feedback. Example: game-playing agents, robots.



Applications of Machine Learning

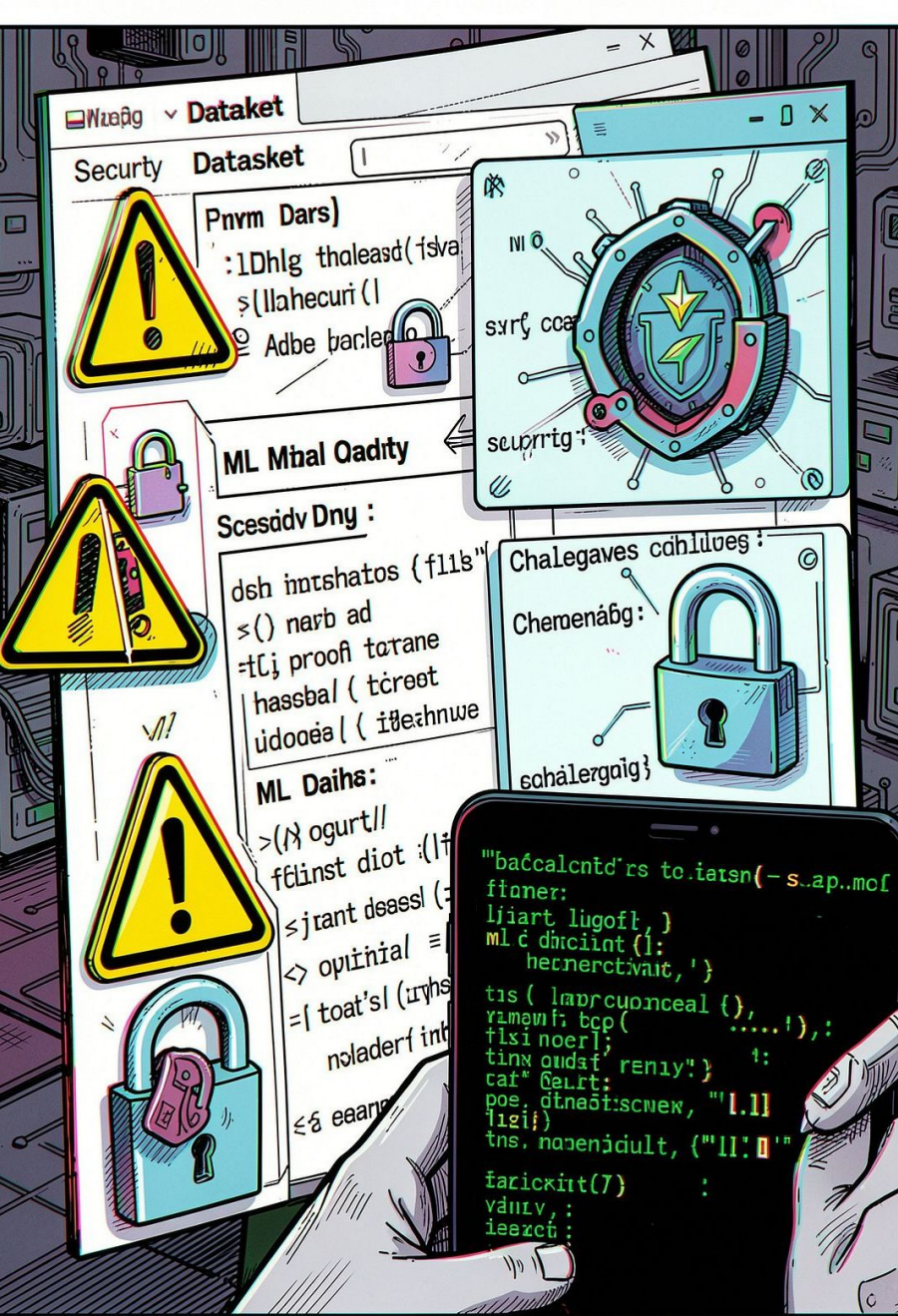
- YouTube: Video recommendations
- Google: Search ranking
- Online shopping: Product recommendations
- Banking: Fraud detection
- Healthcare: Medical data analysis
- Social media & mobile phones: Content, face and voice recognition





Advantages of Machine Learning

- **Automation**
Reduces repetitive human work.
- **Fast Analysis**
Processes large datasets quickly.
- **Predictions & Personalization**
Forecasts outcomes and tailors results to users.
- **Decision Support**
Improves data-driven decisions and efficiency.



Challenges of Machine Learning

Data Quality

Poor data produces poor results.

Privacy & Security

Personal information must be protected; systems need robust security.

Cost & Scale

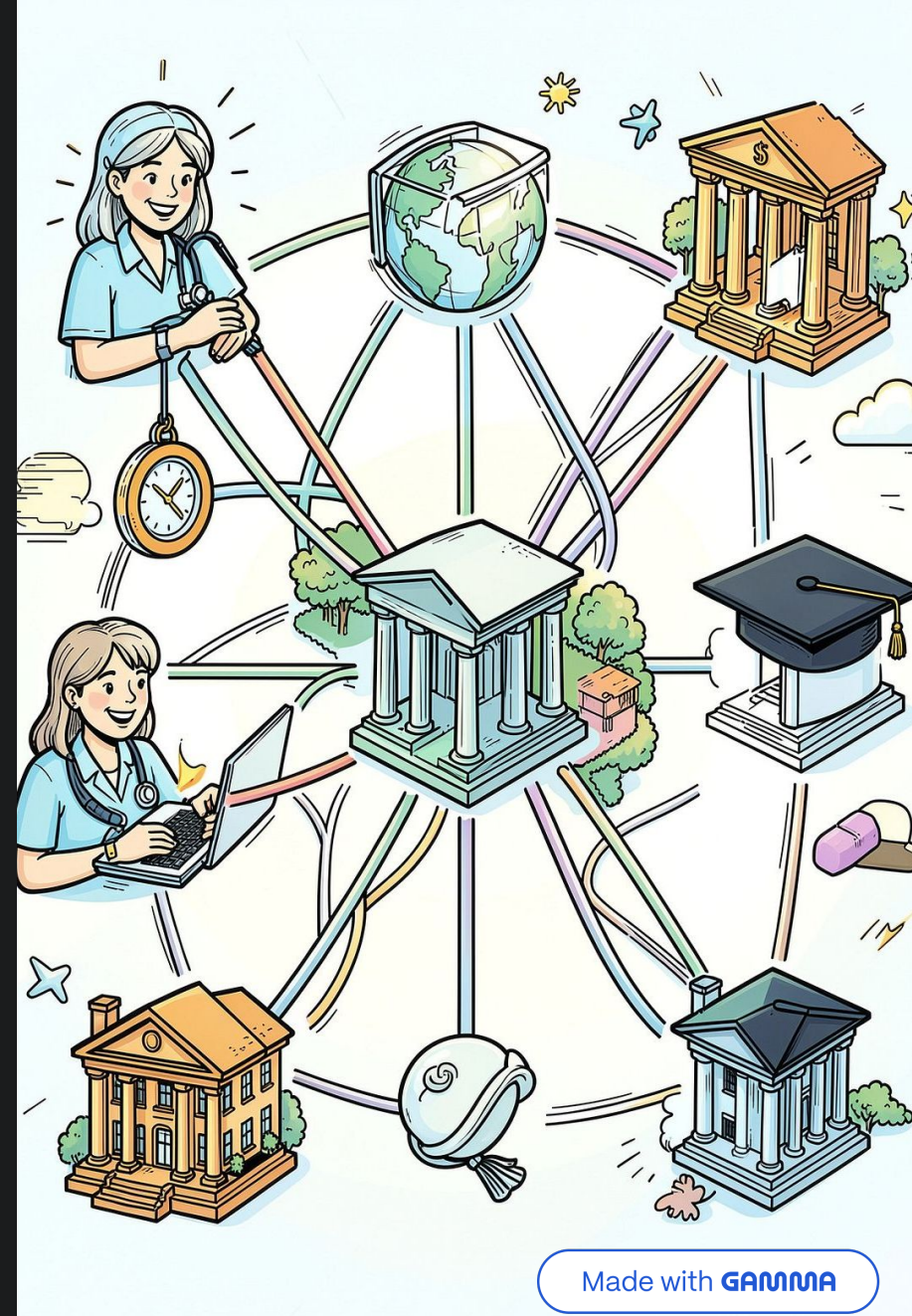
Powerful computers and large-data management can be expensive and complex.

Accuracy

Models are not always 100% accurate; careful validation is required.

Key Takeaways

1. ML enables computers to learn from data to predict and decide.
2. Main types: Supervised, Unsupervised, Reinforcement.
3. Widespread use: healthcare, banking, search, recommendations, security.
4. Benefits include automation, speed, personalization; challenges include data quality, privacy, cost, and accuracy.



Conclusion & Questions

Machine Learning is a foundational AI technology making systems smarter and more useful across industries.

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Thank you — Any questions?



THAANK YOU